Computer Science - K-12 Level Assessment 12 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following has the greatest potential for compromising a user's personal privacy?
   (a) A group of cookies stored by the user's Web browser
   (b) The Internet Protocol (IP) address of the user's computer
   (c) The user's e-mail address
   (d) The user's public key used for encryption

2. What is the value of this python expression: 4 + 3 % 2?
   (a) 1
   (b) 6
   (c) 4
   (d) 5

3. In Python 3, b = [11,13,15,17,19,21]; print(b[::2]) outputs what?

19,21

11,15

11,15,19

13,17,21

4. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
   (a) 17_{16}
   (b) E 4_{16}
   (c) E 7_{16}
   (d) F 4_{16}

5. Which is the smallest asymptotically?
   (a) O(1)
   (b) O(n)
   (c) O(n^2)
   (d) O(log n)

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, floor division is performed using the double forward slash operator '//'.
7. True or False: When someone logs in with a stolen password, it compromises the system's authentication security.
8. True or False: A denial-of-service (DoS) attack targets multiple computers to overwhelm a single system.
9. True or False: In Python 3, the output of print(list[1:3]) with list = ['abcd', 786, 2.23, 'john', 70.2] is [786, 2.23].
10. True or False: Variable names in Python are not case-sensitive and can be used interchangeably regardless of capitalization.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the number of possible sequences of length n such that each of the next element is greater than or equal to twice of the previous element but less than or equal to m.
12. Write a function to perform the adjacent element concatenation in the given tuples.

End of Paper